Sewon Kim

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Objective

Optimal Control and **Reinforcement Learning**. I am interested in developing scalable learning and control methods and their theoretical foundations for robotics.

Education

University of Seoul — *Mar 2022 – Aug 2027 (Expected)*
B.S. in Electrical and Computer Engineering, Minor in Mathematics — Seoul, South Korea
Advisor: Gyunghoon Park, Dohyun Kwon

Experience

Computer Vision AI Engineer at **InBody AI Scale Team, InBody** — *Jan 2024 – Jan 2025*
[B.4] InBody AI Scale — Seoul, South Korea

Undergraduate Intern at **Control and Dynamic System Lab, University of Seoul** — *Mar 2023 – Dec 2024*
[B.3] F1tenth, Advisor: Gyunghoon Park — Seoul, South Korea

Service

Drone Researcher at **Institute of Innovation for Future Army, Republic of Korea Army** — *Jan 2025 – Jul 2026*
Autonomous Drone for Indoor Mapping in Close Quater Battle; Mandatory Military Service — Daejeon, South Korea

Management Team at **AI Robotics KR** — *Sep 2023 – Sep 2024*
Seoul, South Korea

Publications

A.1 **Sewon Kim**, Hyunwoo Lee, Jeongyeol Kwon, and Dohyun Kwon. *First-Order Methods for Trilevel Optimization.* Under review at NeurIPS 2026.

Projects

B.1 **Autonomous air purifying agent** — *Aug 2025 – Sep 2026*
4th Place (/200) ($2,200) — SK NAMUHX A1: Autonomous Navigation Path Optimization
Independently implemented full-stack autonomous air purifying agent in NumPy from scratch, covering MCL, MPPI, and TSP.

B.2 **Autonomous Racing F1tenth** — *Mar 2024 – Dec 2024*
5th Place (/12) — 22nd F1tenth Autonomous Grand Prix at CDC Milan 2024
Led the project with 2 undergrads and implemented a full-stack physical autonomous navigation system for racing from scratch, covering SLAM, MCL, Raceline Opt, MPPI.

B.3 **Cart pole system design and control by PID and NMPC** — *Jul 2024 – Sep 2024*
3rd Place (/25) ($220) — UOS ECE Innovation Fair
Independently developed a physical Cart-Pole system for $272, covering modeling, motor control, sensor integration, state estimation, embedded systems, and MPC.

B.4 **InBody AI Scale** — *Jan 2024 – Jan 2025*
CES Las Vegas 2025
Developed height estimation model (acc 99.4%) and face recognition system of "InBody AI Scale", automated body composition analyzer that reduces measurement time from 120s to 40s.

B.5 **ECG data-driven cardiovascular diagnosis device** — *Jul 2023 – Jul 2023*
3rd Place (/12) ($1450) — InBody Hackathon 2023
Developed a fast prototype.

B.6 **Autonomous mission mobile robot** — *Jul 2023 – Jul 2023*
3rd Place (/20) ($220) — Object Detection-Based Solar Tracking Robot Challenge, Korea University

B.7 **Sidaeting** — *Mar 2022 – Jun 2022*
1,700 User Base
Implemented a matching algorithm that combinatorially optimizes collective happiness from users' ideal type data and developed the frontend for "Sidaeting", a dating service launched by UOSLIFE.

Skills

- **Languages:** Korean, English
- **Technical Languages:** C, C++, Java, JavaScript, Julia, Python (Advanced), SQL, HTML, CSS
- **Technical Frameworks:** Isaac Sim·Lab, ROS1·2 (Melodic, Noetic, Foxy, Humble, Jazzy), MuJoCo, PyTorch (Advanced)